

Consumer Shopping Behavior Analysis

Retail Analytics and Customer Insights

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Data Analytics Project

Python • SQL • PostgreSQL • Power BI

Project Overview

This project analyzes customer shopping behavior using transactional retail data containing approximately 3,900 purchase records across multiple product categories.

The analysis focuses on identifying customer purchasing trends, revenue-driving segments, discount dependency, subscription behavior, and product performance to support business decision-making.

The project follows a complete analytics workflow involving:

- Data preprocessing and feature engineering using Python
- Business analysis using SQL and PostgreSQL
- Dashboard development in Power BI
- Business recommendations based on analytical insights

Dataset Summary

Total Records	3900
Total Features	18
Domain	Retail / E-Commerce
Database	PostgreSQL

Table 1: Dataset Overview

Key attributes included:

- Customer demographics
- Product categories
- Purchase amount
- Discount usage
- Subscription status
- Shipping preferences
- Review ratings

Exploratory Data Analysis using Python

Data preprocessing and exploratory analysis were performed using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn.

Main preprocessing steps included:

- Handling missing values
- Column standardization
- Feature engineering
- Customer segmentation
- Data consistency checks

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN

Figure 1: Initial Dataset Exploration and Summary Statistics

Business Analysis using SQL

The cleaned dataset was loaded into PostgreSQL for business-oriented analysis using SQL queries.

Key analyses included:

- Revenue analysis by gender
- High-spending discount users
- Top-rated products
- Shipping type comparison
- Subscription analysis
- Customer segmentation
- Revenue contribution by age group

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Figure 2: Revenue Analysis by Gender

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:0

Figure 3: High-Spending Discount Users

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Figure 4: Top 5 Products by Average Review Rating

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Figure 5: Average Purchase Amount by Shipping Type

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Figure 6: Subscriber vs Non-Subscriber Revenue Analysis

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Figure 7: Products with Highest Discount Dependency

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Figure 8: Customer Segmentation Analysis

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Figure 9: Top Performing Products by Category

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Figure 10: Repeat Buyers vs Subscription Status

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Figure 11: Revenue Contribution by Age Group

Power BI Dashboard

An interactive dashboard was developed in Power BI to visualize customer behavior trends and business KPIs.

Dashboard highlights included:

- Revenue by category
- Customer segmentation
- Subscription insights
- Revenue by age group
- Sales trends
- Average review ratings



Figure 12: Interactive Customer Shopping Behavior Dashboard

Key Insights

- Male customers generated significantly higher revenue compared to female customers.
- Customers using express shipping showed higher average purchase values.
- Fashion-related products displayed strong discount dependency.
- Loyal customers represented the largest customer segment.
- Young adults contributed the highest revenue among all age groups.

Business Recommendations

- Strengthen customer loyalty programs for repeat buyers.
- Promote subscription memberships through targeted campaigns.
- Optimize discount strategies for high-discount categories.
- Focus marketing efforts on high-revenue customer groups.
- Prioritize top-rated products in advertising campaigns.

END